

Ascot Micro-Forest Connect



Loyola
Marymount
University



WILLETTE LAB OF
APPLIED ECOLOGY

May 1, 2026

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1. Planting Micro-Forests



What is the Ascot Micro-Forest Project?

2. Building Software Infrastructure



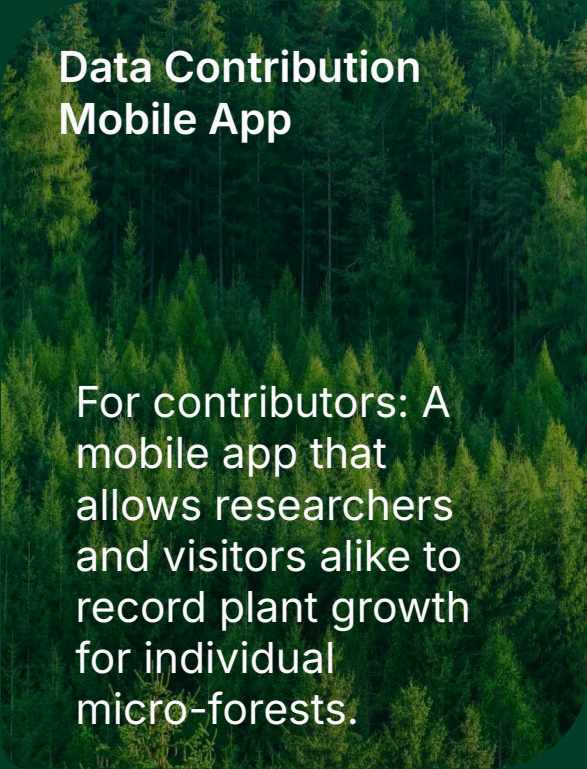
3. Expanding Education and Impact



4. Engaging Communities

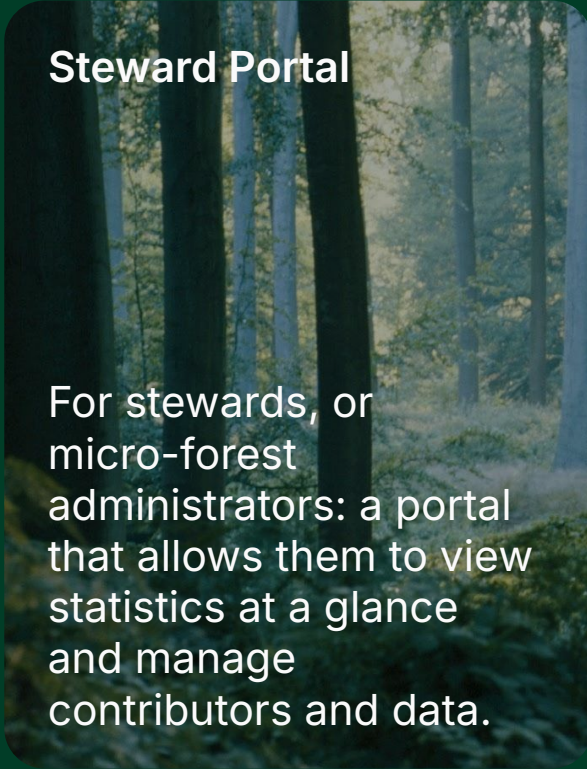


Project Components



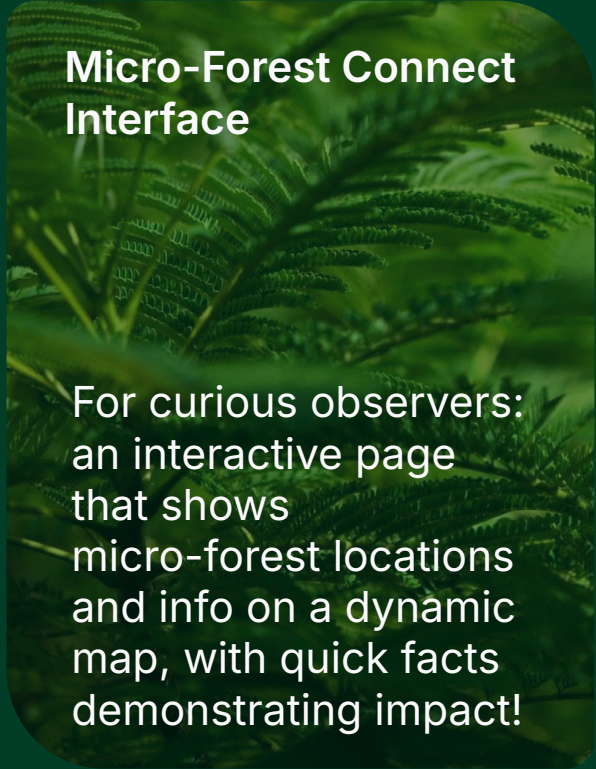
Data Contribution Mobile App

For contributors: A mobile app that allows researchers and visitors alike to record plant growth for individual micro-forests.



Steward Portal

For stewards, or micro-forest administrators: a portal that allows them to view statistics at a glance and manage contributors and data.



Micro-Forest Connect Interface

For curious observers: an interactive page that shows micro-forest locations and info on a dynamic map, with quick facts demonstrating impact!

Project Goals

- **Make contribution easier than ever**
- **Allow new micro-forests to join on**
- **Give stewards control of their data**
- **Improve reach and visibility**
- **Modernize and prepare the project for scale**



Motivation - Why Ascot?

Concept and Subject Matter

- Allowing communities to see how much they're benefiting the environment
- Encourage the proliferation of micro-forests around the LA area and the world

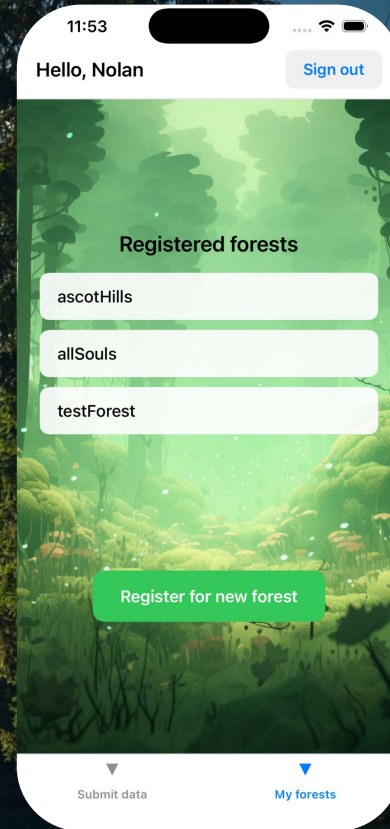
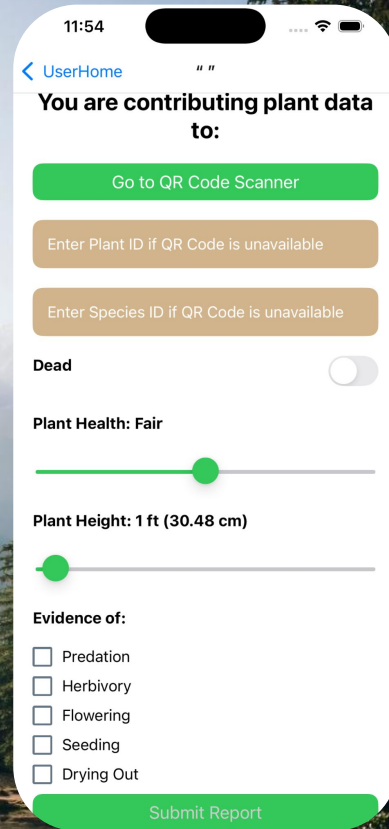
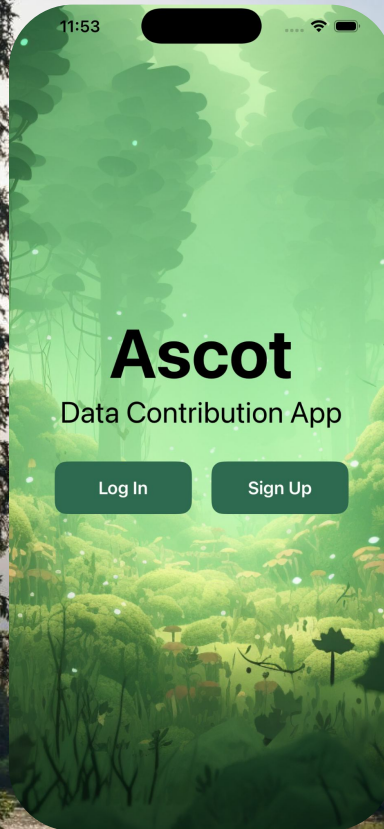
Interdisciplinary Connection

- A chance to collaborate outside of the computer science department
- Combines our technical expertise with the subject matter of another department

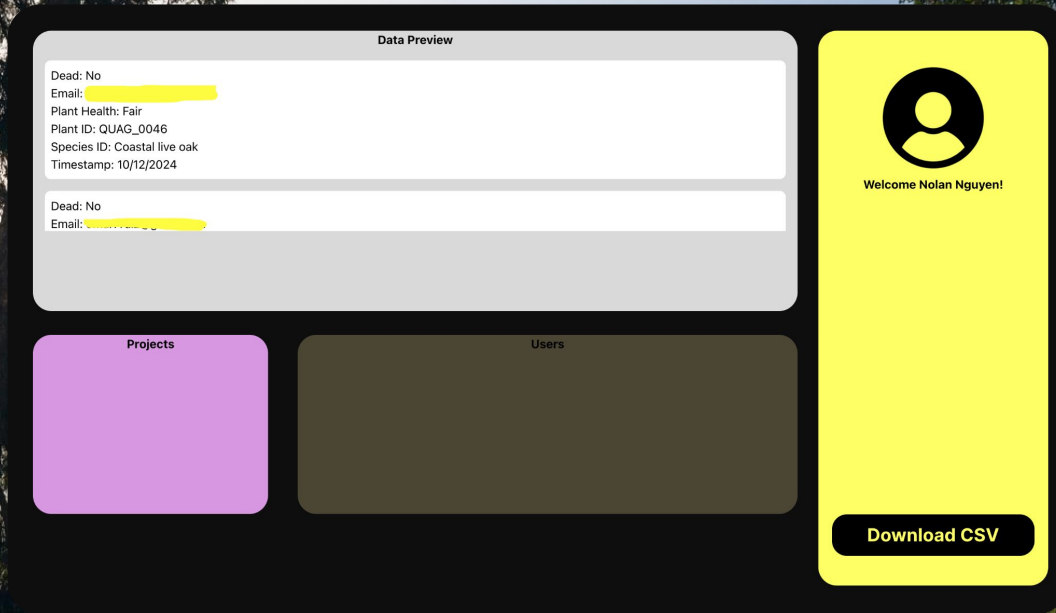
Project and Codebase Format

- Codebase is in a familiar format, we can leverage past experience with full-stack engineering
 - A good chance to exercise communicating with a "customer"/client
-

User Interface

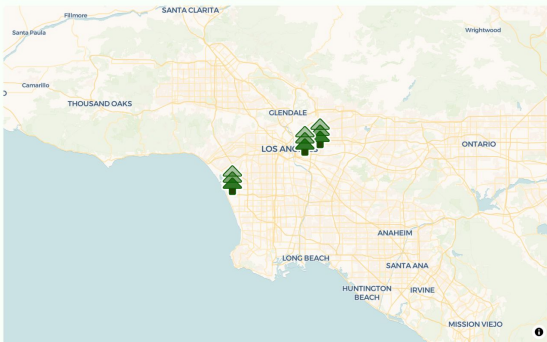


Data Contribution



User Interface

BEFORE: Steward Portal

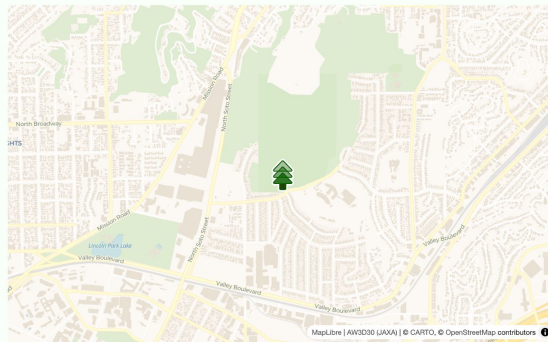


Ascot Hills

Los Angeles, CA

Established 2024 - 10,000 sq ft

This 10,000 square foot micro-forest in LA features 850 native trees, shrubs, and herbaceous perennials planted on a south-facing slope in Ascot Hills Park in Northeast LA. This micro-forest adapts and modifies the Japanese Miyawaki Method of afforestation to create dense, climate-adapted native habitat in Los Angeles, taking lessons learned from the pilot micro-forest planted in Griffith Park in 2021. All plants were grown from locally-sourced seeds in partnership with Grown in LA and North East Trees. Biodiversity, carbon, and survivorship research is being conducted by students and faculty at Loyola Marymount University. At only 18 months old, the micro-forest is already supporting 50% more biodiversity than an adjacent control plot, and is sequestering one ton of carbon annually, with an estimation of 55 tons of carbon sequestered per year after 20 years.



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User Interface

Micro-Forest Connect



..Back

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About

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Location

4201 Mulholland St, Los Angeles, CA 90032

Coordinates: 34.07305, -118.19887

Open in Google Maps

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Micro-Forest Connect is supported by



Demonstration



Project Challenges

Managing 3 Different Types of Users

- Ascot involves 3 user bases with different priorities: Contributor, Steward, Observer
- Decisions on security, accessibility, features, and more are decided separately per-component

Metric Generation from Raw Data

- We want to calculate complex statistics like CO₂ absorption from our data
- But, have to start with rudimentary data points like plant height, diameter, and health indicators

Codebase Modernization

- The original codebase was in need of updating, to new versions and new technologies altogether
 - All three components use different developer libraries to be deployed and hosted, while all hooking up to the same database
-

Status Update & Punch List

Data Contribution App

- The Data Contribution App is already online and ready to serve!
 - New update will allow plants to be tracked in separate databases representing different micro-forests, to prepare for scale in the Ascot Project
 - Forests can be joined using a verification code
-

Steward Portal

- Steward Portal has been completely overhauled since beginning the project, from a web app stub to an informative dashboard for micro-forest managers
 - Currently working on user flow for creating a new micro-forest, allowing Dr. Willette to approve requests to join the project
-

Micro-Forest Connect

- The Micro-Forest Connect Interface is the newest component in the project
 - Currently has an interactive map showing forest locations and quick facts
 - Working on compiling plant data into statistics that track impact, such as CO2 absorption rate
-

Punch List:

- Finish Create New Micro-Forest flow so that Dr. Willette can approve instantiation of new micro-forests
- Make blog/editorial components of Micro-Forest Connect editable by Dr. Willette without having to code
- Ascot will continue to evolve and grow – write up documentation and jumping-off points for future devs

Thank you! And thank you to:

Dr. Willette
Professor Dondi
Professor BJ

Christina Choi '24
Coby Sanders '24
Aaron Floreani '24
Evan Yu '24

Any questions?



For any further questions,
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